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한국에너지공과대학교

붙임1

참여 연구실 및 연구 주제 소개

연번	트랙	지도교수	연구실명	Research Keyword ※ 상세 내용은(https://gs.kentech.ac.kr/composition/) 참고
1	에너지 AI	김종권	TEAT (A Technique to ...)	Social network analysis, Reinforcement learning, IoT, Graph neural network, Recommender system, Video streaming
2	에너지 AI	신욱철	Robust Physical AI & Robotics Lab (RoBust Lab)	Physical AI, Robot Learning, Reinforcement Learning, Robotics, Robot Vision, 3D geometry, Sensor Fusion, Vision-Language Robot Control
3	에너지 AI	안수명	Robust Energy AI Lab (REAL)	Machine learning / Artificial Intelligence: Robust and Trustworthy AI, Foundation Models (NLP, Multi-modal models)
4	에너지 AI	이범식	ENergy, Vision Intelligence Lab (ENVI Lab)	Image and Video Processing, Computer Vision, Artificial Intelligence for Energy, Medical Imaging and Analysis
5	에너지 AI	이석주	Visual Intelligence Energy-Wise (Viewlab)	Computer Vision; Machine Learning; Robotics; Autonomous Driving; Visual Perception; 3D Vision; Sensor Fusion
6	에너지 AI	이현우	Computer System Security Lab (CSSLab)	Network Security, Applied Cryptography, Protocol Verification, Intrusion Detection, Trusted Execution Environment, Internet of Things, AI for Security
7	에너지 AI	임혁	Sustainable AI Lab	Machine learning and artificial intelligence Big data privacy and AI security AI application for cybersecurity, networks, and energy
8	에너지 신소재	송주현	Battery Software Group	Li-ion battery, battery management system (BMS), electrochemical modeling, multi-scaling modeling, multi-physics modeling, algorithm optimization, techno-economic evaluation
9	에너지 신소재	오상호	Electron Microscopy for Energy Materials via Environment Manipulation (eM3)	In-situ TEM/STEM, Electron holography/ptychography, Deformation mechanisms, Interface/surface structure, Switching mechanisms of devices, Strain mapping, Electron imaging
10	에너지 신소재	윤재호	NextGen Photovoltaic Materials & Devices Lab (PMDL)	Thin Film Solar cells : Chalcogenide, earth-abundant materials, CIGS, Flexible, color and semi-transparent solar cells Tandem & multi-junction solar cells :SI-based tandem cell, CIGS-based tandem cell, Perovskite vacuum deposition, Multi-junction cell Multi-functional PV modules : BIPV(Building Integrated PV), V(Vehicle)IPV, R(Road)IPV PV-based hybrid energy system : Energy sharing platform, Sector coupling
11	차세대 그리드	김희태	Grid Complexity Lab	Network science, power grid, synchronization stability, community detection, sustainability, greenhouse gas emissions
12	차세대 그리드	박화평	AI-PEL	Grid connected inverters and DC/DC converters for renewable energies, Fault detection and extinguishing technologies for power electronics, Reliability and safety improvement for battery related systems, Control algorithms for power conversion efficiency optimization, Power electronics for industrial applications
13	차세대 그리드	이진오	Power Systems Laboratory	DC Power Systems, Hybrid AC/DC Microgrids, Inverter-Dominated Power System, Power System Stability

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14	수소 에너지	구근호	Meta Lab	Artificial Intelligence, Catalysis and Materials Science, Multiscale Modeling, Computational Chemistry
15	환경 기후 기술	김우열	Operando analysis for photo/electro-catalysis lab	Energy application-Solar fuel (artificial photosynthesis), Green ammonia & Water splitting Environmental applications-Wastewater & Air pollutant removal Time-resolved spectroscopy(From UV-visible to IR)
16	환경 기후 기술	오명환	Artificial Living Systems (ALS)	Energy nanoscience, Materials synthesis, Heterogeneous catalysis, Electrochemical system, Artificial photosynthesis, Artificial living systems